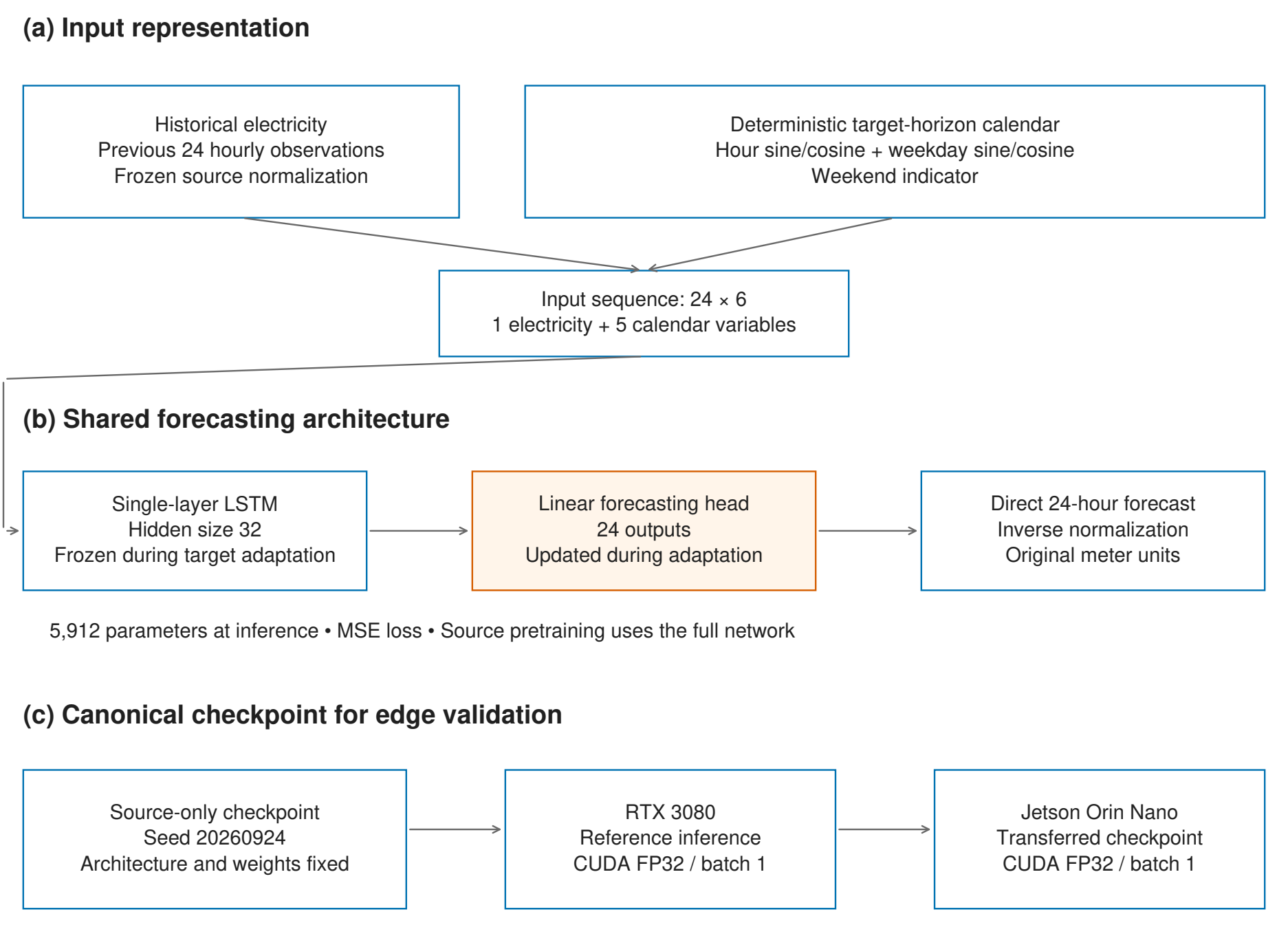
(a) Input representation
Historical electricity
Previous 24 hourly observations
Frozen source normalization
Deterministic target-horizon calendar
Hour sine/cosine + weekday sine/cosine
Weekend indicator
Input sequence: 24 × 6
1 electricity + 5 calendar variables
(b) Shared forecasting architecture
Single-layer LSTM
Hidden size 32
Frozen during target adaptation
Linear forecasting head
24 outputs
Updated during adaptation
Direct 24-hour forecast
Inverse normalization
Original meter units
5,912 parameters at inference • MSE loss • Source pretraining uses the full network
(c) Canonical checkpoint for edge validation
Source-only checkpoint
Seed 20260924
Architecture and weights fixed
RTX 3080
Reference inference
CUDA FP32 / batch 1
Jetson Orin Nano
Transferred checkpoint
CUDA FP32 / batch 1
Validation: numerical consistency and inference cost for the shared architecture.
This experiment does not establish the accuracy of every adapted checkpoint.